

Errata

Physical Geodesy: A Theoretical Introduction

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1. Page 3, lines 8 to 9:

Change “as attraction, of m ” to “as attraction of m ”

2. Page 5. Fig. 1.3 caption:

Change “acceleration” to “attraction”

3. Page 26. Below eqn (1.85) and below eqn (1.88):

Change “disc” to “disk”

4. Page 40, below eqn. (1.151):

$$(\Delta x^2 + \Delta z^2)^{1/2} = [1 + (dz/dx)^2]^{1/2} \Delta x$$

5. Page 89. eqn (3.45):

$$\delta = \lim_{i \rightarrow \infty} \left| \frac{C_i}{C_{i+2}} \right| = \lim_{i \rightarrow \infty} \left| \frac{(i+2)(i+1)}{i(i+1) - \alpha} \right| = 1.$$

6. Page 91, eqn (3.58):

$$y_1^0(x) = \sum_{i=1}^{\infty} \frac{1}{2i} x^{2i} \quad \text{and} \quad y_2^0(x) = \sum_{i=0}^{\infty} \frac{1}{2i+1} x^{2i+1}$$

7. Page 91, add to the end of the paragraph of (3.58):

The factor M in (3.57) is dropped, since it does not affect the convergence.

8. Page 91, eqn (3.60):

$$\left\{ \begin{array}{l} \frac{1}{2} [\ln(1+x) + \ln(1-x)] = - \sum_{i=1}^{\infty} \frac{x^{2i}}{2i}, \\ \frac{1}{2} [\ln(1+x) - \ln(1-x)] = \sum_{i=0}^{\infty} \frac{x^{2i+1}}{2i+1}. \end{array} \right.$$

9. Page 105, eqn (3.132):

$$P_n^k(\pm 1) = 0, \quad k \neq 0;$$

10. Page 105, eqn (3.134):

Replace the last “.” by “,”

11. Page 131, eqn (3.270):

Replace “k=1” by “k=0” under the summation sign.

12. Page 136, in equation (3.285):

Change “ C_1^0 ” to “ C_1 ”

13. Page 142, eqn (3.314):

$$V = \frac{GM}{r} \sum_{n=0}^{\infty} \left(\frac{a}{r}\right)^n \sum_{k=0}^n (\bar{C}_n^k \cos k\lambda + \bar{S}_n^k \sin k\lambda) \bar{P}_n^k(\cos \theta),$$

Remark: Including the degree 1 normalized coefficients is a better choice.

14. Page 144, eqn (3.321):

Replace “k=1” by “k=0” under the summation sign.

15. Page 144, eqn (3.322):

Replace “k=1” by “k=0” under the summation sign.

16. Page 145, eqn (3.327 should be):

$$\sum_{k=0}^n \bar{P}_n^k(\cos \theta) \frac{d}{d\theta} \bar{P}_n^k(\cos \theta) = 0.$$

17. Page 153, below eqn (3.370):

Change “Here we ... be written as” to “On the equator, with $\theta = \pi/2$, this formula can be written as”

18. Page 159, last sentence of the Abstract:

Change “called Geodetic Reference System (GRS)” to “which are essential for a Geodetic Reference System (GRS)”

19. Page 199, line 7 from bottom:

Change “which are referred to as Geodetic Reference System” to “which are essential for a Geodetic Reference System”

20. Page 200, the line above eqn (4.180):

Change “are the Geodetic Reference System GRS80” to “are those of the Geodetic Reference System 1980 (GRS80)”

21. Page 202, line 7 from bottom, the reference should be replaced by:

Borkowski, K.M. (1989). Accurate algorithms to transform geocentric to geodetic coordinates. Bulletin Géodésique, 63, 50-56.

22. Page 204, the first line of the paragraph above the Section title 5.1.2:
Change “defers” to “differs”

23. Page 209, eqn (5.20):

$$\left(\frac{\partial U}{\partial n_{\Sigma}} \right)_{\Sigma} = \left(\frac{\partial U}{\partial n_S} \right)_S + N \left(\frac{\partial^2 U}{\partial n_S^2} \right)_S = \left(\frac{\partial U}{\partial n_S} \right)_S - N \left(\frac{\partial \gamma}{\partial n_S} \right)_S .$$

24. Page 246, Title of Sect. 5.5.2:

Change “Degree Power Spectrum” to “Degree Variance Power Spectrum”

25. Page 246, Title of Fig. 5.10:

Change “Degree power spectrum variance” to “Degree Variance Power Spectrum”

26. Page 247, eqn (5.172):

$$V_n = \frac{GM}{R} \left(\frac{R}{r} \right)^{n+1} \sum_{k=0}^n \left[\frac{2(2n+1)}{1+\delta_k} \frac{(n-k)!}{(n+k)!} \right]^{1/2} \\ \times (\bar{C}_n^k \cos k\lambda + \bar{S}_n^k \sin k\lambda) P_n^k(\cos \theta) ,$$

27. Page 247, eqn (5.174):

$$\frac{1}{4\pi} \int_{\omega} V^2 d\omega = \left(\frac{GM}{R} \right)^2 \sum_{n=0}^{n_{\max}} \left(\frac{R}{r} \right)^{2n+2} \sum_{k=0}^n \left[\left(\bar{C}_n^k \right)^2 + \left(\bar{S}_n^k \right)^2 \right] \\ = \sum_{n=0}^{n_{\max}} \left(\frac{1}{4\pi} \int_{\omega} (V_n)^2 d\omega \right) .$$

28. Page 247, line 3 from bottom:

Delete the sentence “The correction $\Delta_1 g + \Delta_2 g$ is referred to as incomplete Bouguer correction”

29. Page 247, Last line:

Change “Bouguer correction and anomaly” to “Bouguer anomaly”

30. Page 258, eqn (6.13):

$$\Delta_3 g = G\rho \int_0^{2\pi} d\alpha \int_0^a dr \int_0^{\Delta H} \frac{zr}{(r^2 + z^2)^{3/2}} dz ,$$

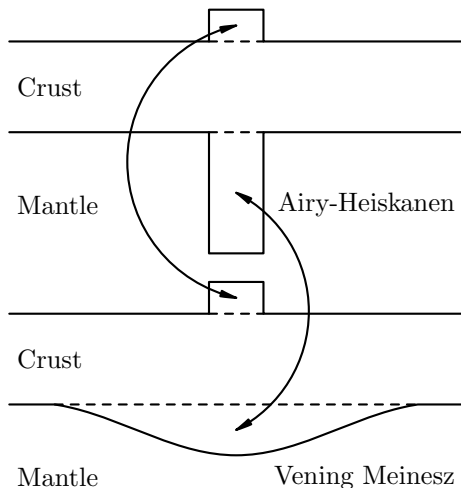
31. Page 258:

Change the sentence above (6.15) to: “The gravity anomaly computed according to”

32. Page 263, eqn (6.27):

$$\Delta\rho = \rho - \rho' = -\frac{(\rho - \rho_w)P}{D - P}.$$

33. Page 265, Fig. 6.7 (It makes me feel embarrassed to have the one in the print! I went to replace it before the submission of the final version to Springer, but I have mistakenly replaced it in a different draft):



34. Page 301, Line 5:

Change ”~ 10 m for the isostatic anomaly.” to ”~ 10 m for the isostatic anomaly (Heiskanen and Moritz, 1967).”

35. Page 313, eqn (6.223):

$$f_T(T, H', \psi) = \frac{R^2}{2} \left\{ (x + 3 \cos \psi)(1 + x^2 - 2x \cos \psi)^{1/2} + (3 \cos^2 \psi - 1) \right. \\ \left. \times \ln \left[(1 + x^2 - 2x \cos \psi)^{1/2} + x - \cos \psi \right] \right\}^{1-T/R}_{1-(T+H')/R}.$$

36. Page 319:

Add the following into the references:

Heiskanen, W.A., & Moritz, H. (1967). *Physical geodesy*. W.H. Freeman and Company.

37. Page 393 line 3 and page 493 line 7 from bottom (So embarrassed to misspell a colleague’s name):

Change “Bgherbandi” to “Bagherbandi”.

The same correction applies to Page 493, line 7 from the bottom.

38. Page 398, line 26 (or line 9 of the third paragraph):

Change “If not, the difference between the differences” to “If not, the sum of the two differences”

39. Page 410, line 6:

Change “ $n_{\max}(n_{\max} + 1)$ ” to “ $(n_{\max} + 1)^2$ ”

40. Page 415, line 2 of the last paragraph:

Add “, although the decrease may be oscillatory” after “ β decreases with n ”

41. Page 417, line 3 from bottom:

Change “ $\int_{-\sigma}^{\sigma} g(x)dx = 99.73\%$ ” to “ $\int_{-3\sigma}^{3\sigma} g(x)dx = 99.73\%$ ”

42. Page 419:

Add a “,” after (8.75) just above eqn (8.78).

43. Page 423: Eqn (8.98) should be

$$\sum_{i=1}^N \sum_{j=1}^{2N} W_{ij} \left[\sum_{n=2}^{N-1} \sum_{k=0}^n (\alpha_{ijnk} \bar{c}_n^k + \beta_{ijnk} \bar{s}_n^k) - \Delta \bar{g}_{ij} \right]^2 = \min$$

44. Page 424:

Change the paragraph starting the 3rd line to

“The second problem is the resolvability. In this approach, the coefficients $\bar{c}_n^k$ and $\bar{s}_n^k$ can be reliably solved only up to degree and order $N - 1$ (Colombo, 1981). This could be demonstrated by examining the problem on the equator where $\theta = \pi/2$ and (8.32) degenerates to the Fourier series

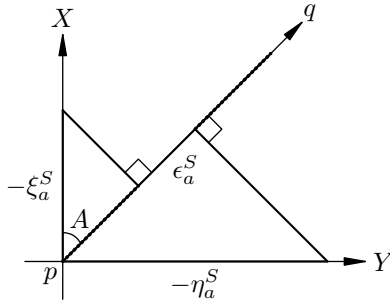
$$\Delta g(\lambda) = \sum_{k=0}^{n_{\max}} (a_k \cos k\lambda + b_k \sin k\lambda). \quad (8.99)$$

Hence, there are $2n_{\max} + 1$ coefficients to be determined. However, there are $2N$ values of Δg on the equator. The total number of coefficients to be determined cannot be more than the total number of data. As a result, the coefficients can only be solved up to order $n_{\max} = N - 1$ at most. With $n_{\max} = N - 1$ as chosen in (8.98), the total number of coefficients $\bar{c}_n^k$ and $\bar{s}_n^k$ is N^2 , while the total number of data $\Delta \bar{g}_{ij}$ is $2N^2$.”

45. Page 438, second line below 8.5.1:

Change “base” to “based”

46. Page 439, Fig. 8.13:



47. Page 451. eqn (9.14):

Change " $\frac{d}{\partial r}$ " to " $\frac{d}{dr}$ "

48. Page 452. eqn (9.19):

Change " $Y_n^k(\theta, \lambda)$ " to " $Y_n(\theta, \lambda)$ "

49. Page 461, the line above (9.66):

Change "We now assume" to "We now make the finer approximations"

50. Page 461, the line below (9.66):

Change "this assumption" to "these approximations"